

The Job Sequencing and Tool Switching Problem

Part VII: SSP Variants that Collapse to JGP

Scope and Status. This document addresses the question: can we design a variant of the SSP with a modified cost function f such that solving the variant *collapses* to solving the JGP? We present: (i) a formal definition of collapse; (ii) the TSIP as a known baseline case (**proved in literature**); (iii) **sufficient conditions** for collapse (C1+C2+C3) — **proved**; (iv) the magazine setup time variant S_{stop} and when it causes collapse — **proved**; (v) the necessity of the conditions — **open problem (TODO)**. The document explicitly distinguishes proved from open results.

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What Does “Collapse to JGP” Mean?

Definition 1.1 (SSP- f variant). Given a cost function $f : \mathcal{C} \times \mathcal{C} \rightarrow \mathbb{R}_{\geq 0}$, the SSP- f variant minimises $\sum_k f(C_k, C_{k+1})$ over all feasible job sequences and magazine configurations.

Definition 1.2 (Collapse to JGP). SSP- f *collapses to JGP* if for every instance I :

$$\text{OPT}_{\text{SSP-}f}(I) \text{ is achieved by a solution using exactly } Z_{\text{JGP}}^*(I) \text{ groups.}$$

Equivalently: using more than Z_{JGP}^* groups never strictly reduces the SSP- f objective.

If collapse holds, the JGP can be solved first and the SSP- f objective minimised within those groups, dramatically simplifying the problem.

Baseline: The Tool Switching Instants Problem (TSIP)

Definition 2.1 (TSIP cost function). $f_{01}(C_k, C_{k+1}) = \mathbf{1}[C_k \neq C_{k+1}]$: the cost is 1 if the configuration changes, 0 otherwise.

The TSIP minimises the total number of *configuration changes* (regardless of how many tools are swapped each time). With K groups, there are at most $K - 1$ configuration changes, so the TSIP objective is $\leq K - 1$.

Theorem 2.2 (TSIP collapses to JGP [?]). *Under f_{01} , the optimal SSP- f_{01} solution uses exactly Z_{JGP}^* groups.*

Proof. Any solution with K configuration changes corresponds to processing K groups (one per distinct configuration used). So $\text{OPT}_{\text{TSIP}} = Z_{\text{JGP}}^* - 1$ (minimum configuration changes = minimum groups minus one). $\square$ $\square$

Remark 2.3. The TSIP is solvable in polynomial time when tool loading/unloading is unconstrained [?]. It models situations where the machine must stop completely for each tool change (the “batch” model).

Sufficient Conditions for Collapse (Proved)

We give general conditions under which a cost function f causes SSP- f to collapse to JGP.

Theorem 3.1 (Sufficient conditions for collapse). *SSP- f collapses to JGP if f satisfies all three of the following conditions:*

(C1) Self-cost zero: $f(C, C) = 0$ for all $C \in \mathcal{C}$.

(C2) Strict inter-group positivity: *There exists $\delta > 0$ such that for any JGP-optimal grouping $\mathcal{G}^*$, any two groups $G_k \neq G_{k'}$, and any valid configurations $C \in \mathcal{C}(G_k)$, $C' \in \mathcal{C}(G_{k'})$: $f(C, C') \geq \delta > 0$.*

(C3) No-beneficial-split: *For every feasible grouping $\mathcal{G}$ with $K > K^*$ groups, the SSP- f cost for any solution over $\mathcal{G}$ is $\geq$ the SSP- f cost for the best K^* -group solution.*

Proof. Under (C2), any solution with K groups incurs $\geq K - 1$ inter-group transitions, each costing $\geq \delta > 0$, so total cost $\geq (K - 1)\delta$. For $K > K^*$: $(K - 1)\delta > (K^* - 1)\delta$, meaning a K^* -group solution with cost $(K^* - 1)\delta$ (e.g., by using the TSIP baseline) is strictly cheaper than any $(K + 1)$ -group solution. Condition (C3) ensures that no splitting of groups into more pieces can further reduce cost. Together, every optimal solution under SSP- f uses exactly K^* groups. $\square$ $\square$

Remark 3.2 (Necessity — open problem). The sufficient conditions (C1)+(C2)+(C3) are *not* proved to be necessary. It is possible that f causes collapse even when (C2) fails (e.g., if some inter-group transitions have zero cost but the overall structure prevents benefiting from more groups). Proving necessity, or finding a weaker necessary and sufficient condition, is an open problem; see ??.

The Magazine Setup Time Variant (Proved)

Model

In many FMS environments, changing any tool requires stopping the machine. This stop incurs a fixed overhead S_{stop} (machine re-initialisation, operator time) in addition to the per-tool switching cost c_{switch} . The total cost of a transition from C_k to C_{k+1} is:

$$f_S(C_k, C_{k+1}) = S_{\text{stop}} \cdot \mathbf{1}[C_k \neq C_{k+1}] + c_{\text{switch}} \cdot |C_{k+1} \setminus C_k|.$$

When $S_{\text{stop}} = 0$ this reduces to the standard U-SSP. When $c_{\text{switch}} = 0$ (no individual switch cost) this is the TSIP (Section 2). The interesting regime is $S_{\text{stop}} > 0$ and $c_{\text{switch}} > 0$.

Collapse Condition

Theorem 4.1 (Setup-time collapse to JGP). *If the magazine setup time satisfies*

$$S_{\text{stop}} > n \cdot b \cdot c_{\text{switch}},$$

then SSP- f_S collapses to JGP: the optimal solution uses exactly Z_{JGP}^ groups.*

Proof. Consider two competing solutions: one with K groups (configuration changes) and one with $K + 1$ groups.

The maximum possible total *per-tool* switching cost across the entire sequence of n jobs and b -capacity magazine is bounded by $n \cdot b \cdot c_{\text{switch}}$ (at most b insertions per job, n jobs total).

So any solution with K groups costs at most $K \cdot S_{\text{stop}} + n \cdot b \cdot c_{\text{switch}}$. Any solution with $K + 1$ groups costs at least $(K + 1) \cdot S_{\text{stop}}$.

Under the condition $S_{\text{stop}} > n \cdot b \cdot c_{\text{switch}}$:

$$(K + 1)S_{\text{stop}} > K \cdot S_{\text{stop}} + n \cdot b \cdot c_{\text{switch}} \geq \text{cost of any } K\text{-group solution.}$$

Therefore, no $(K + 1)$ -group solution can beat the best K -group solution. Since this holds for any K , the optimal number of groups is minimised — exactly the JGP objective. $\square$ $\square$

Remark 4.2 (Interpretation). When the setup time is large enough to dominate the per-tool cost of the *entire* sequence, it is always better to minimise stops (groups) than to minimise individual tool insertions. The boundary $S_{\text{stop}} = n \cdot b \cdot c_{\text{switch}}$ is the *collapse threshold*. Below this threshold, the SSP and JGP optima diverge.

Parameterised Collapse: The ρ -Frontier

Status: Proposed research direction.

Define the ratio $\rho = S_{\text{stop}}/c_{\text{switch}}$. As ρ increases from 0 to ∞ :

- $\rho = 0$: pure U-SSP; JGP and SSP optima can differ by up to $Z_{\text{JGP}}^* - 1$ switches (Theorem 2 of 05_jgp_ssp_gap_analysis.tex).
- $\rho > nb$: complete collapse (Theorem 4.1).
- $0 < \rho < nb$: partial regime where the optimal number of groups is between Z_{JGP}^* and some $K_\rho > Z_{\text{JGP}}^*$.

Proposed direction. Characterise the exact threshold ρ_c at which the optimal number of groups transitions from Z_{JGP}^* to $Z_{\text{JGP}}^* + 1, Z_{\text{JGP}}^* + 2, \dots$. This would give a complete picture of how the JGP/SSP tradeoff depends on the physical parameters of the machine. This is related to the Lagrangian relaxation of 06_grouping_selection.tex, where $\lambda \approx \rho$ penalises extra groups.

The TD-SSP and Lexicographic Collapse

Tool-Dependent SSP (TD-SSP)

In the *Tool-Dependent SSP* (TD-SSP), the cost of inserting tool t is $c_t > 0$ (not necessarily uniform). The transition cost is:

$$f_{\text{TD}}(C_k, C_{k+1}) = \sum_{t \in C_{k+1} \setminus C_k} c_t.$$

When $c_t = 1$ for all t , this reduces to the U-SSP. When c_t are arbitrary, the KTNS policy is no longer optimal (it assumes uniform costs). However, the Tooling Problem (TP) for a fixed sequence remains polynomial via minimum-cost network flow on an interval graph (totally unimodular constraint matrix) [?].

Lexicographic Collapse Under Heavy Penalties

When a machine stop incurs a *fixed* penalty S_{stop} in the TD-SSP model, the total cost is:

$$Z_{\text{TD}} = K \cdot S_{\text{stop}} + \sum_{k=1}^{K-1} f_{\text{TD}}(C_k, C_{k+1}).$$

Theorem 5.1 (Lexicographic JGP recovery for TD-SSP). *If $S_{\text{stop}} > n \cdot b \cdot \max_t c_t$, the TD-SSP objective is lexicographically: first minimise the number of groups K (JGP objective), then minimise the total tool-switching cost within those groups.*

Proof. The maximum possible total tool-switching cost (second objective) across the entire sequence is bounded by $\sum_k \sum_{t \in C_{k+1} \setminus C_k} c_t \leq n \cdot b \cdot \max_t c_t$. Under the stated condition, this entire sum is strictly less than S_{stop} . Therefore, adding one extra machine stop (increasing K by 1) costs at least S_{stop} , which strictly dominates any possible savings in the per-tool cost. The primary objective is therefore to minimise K (JGP), and the secondary objective (per-tool cost) is resolved within the Z_{JGP}^* -group structure. $\square$ $\square$

Remark 5.2 (Practical relevance). This theorem justifies the use of the JGP as a primary objective in settings where machine downtime is very costly (e.g., aerospace manufacturing,

high-precision machining). In such settings, the JGP+GSP heuristic is *exactly optimal* for the modified objective, not merely an approximation.

Analysis of Collapse Conditions

Which Conditions are Necessary?

The conditions (C1)+(C2)+(C3) in Theorem 3.1 are *sufficient* for collapse, but not all are necessary.

Proposition 6.1 (Necessity analysis). *Of the three conditions (C1), (C2), (C3) in Theorem 3.1:*

- (i) (C3) is necessary by definition of collapse.
- (ii) (C1) is not necessary in general.
- (iii) (C2) is not necessary in general.

Proof. (C3) necessary. If there exists a grouping with $K > K^*$ groups whose SSP- f cost is strictly lower than all K^* -group solutions, collapse fails by definition. So (C3) is necessary.

(C1) not necessary. Let $f(C, C') = \varepsilon > 0$ for all pairs including $C = C'$ (positive self-cost). Any solution with K groups pays $(K - 1)\varepsilon$ regardless of self-costs. Minimising K still gives collapse, so (C1) fails (self-cost positive) yet collapse holds.

(C2) not necessary. Take $J = \{j_1, j_2, j_3\}$, $b = 3$, $T_1 = \{t_1, t_2, t_3\}$, $T_2 = \{t_3, t_4, t_5\}$, $T_3 = \{t_1, t_4, t_6\}$. Here $K^* = 3$ (each pair requires > 3 tools). Define: $f(C_1, C_2) = 0$, $f(C_2, C_3) = f(C_1, C_3) = 1$, $f(C, C) = 0$. The best 3-group solution costs $f(C_1, C_2) + f(C_2, C_3) = 0 + 1 = 1$. No 4-group solution achieves cost < 1 (it still must cross the boundary between some pair with positive cost). Collapse holds despite $f(C_1, C_2) = 0$ violating (C2). $\square$ $\square$

Remark 6.2 (Minimal sufficient condition). (C3) is the only necessary condition, but it is tautological (it states the definition of collapse). The conditions (C1)+(C2)+(C3) form a clean and *verifiable* sufficient criterion. Finding a tight non-trivial necessary-and-sufficient characterisation of all f for which SSP- f collapses to JGP remains open.

The Collapse Threshold $\rho_c(I)$

For the setup-time model f_S with ratio $\rho = S_{\text{stop}}/c_{\text{switch}}$, the exact collapse threshold is computable per instance.

Theorem 6.3 (Collapse threshold formula). *Define the switch-savings function:*

$$\Delta(I, K) = \min_{K^* \text{-group solutions}} \text{cost}_{c=1}(\cdot) - \min_{K \text{-group solutions}} \text{cost}_{c=1}(\cdot), \quad K > K^*.$$

The collapse threshold is $\rho_c(I) = \sup_{K > K^} \Delta(I, K)/(K - K^*)$. SSP- f_S collapses to JGP if and only if $\rho > \rho_c(I)$.*

Proof. A K -group solution costs $(K - 1)S_{\text{stop}} + c_{\text{switch}} \cdot z_K$. It beats the K^* -group optimal iff $c_{\text{switch}}(z_{K^*} - z_K) > (K - K^*)S_{\text{stop}}$, i.e. $\rho < \Delta(I, K)/(K - K^*)$. Collapse fails iff some K

satisfies this; collapse holds iff $\rho \geq \rho_c(I)$. □ □

Remark 6.4 (The 6-job ring). For the ring instance (Part V, Example 1): $K^* = 3$, best 3-group GSP cost $z_3 = 4$, SSP optimum $z_4 = 3$ with 4 groups. So $\rho_c(\text{ring}) = \Delta(I, 4)/(4-3) = (4-3)/1 = 1$. For $\rho > 1$: JGP is optimal (TSIP regime). For $\rho \leq 1$: 4 configurations may be beneficial.

Summary

Variant	Cost f	Collapse?	Status
U-SSP	$ C_{k+1} \setminus C_k $	Not in general	Counterex. in Part V
TSIP	$\mathbf{1}[C_k \neq C_{k+1}]$	Yes, always	Proved
$f_S, S_{\text{stop}} > nbc$	Fixed stop + per-tool	Yes	Proved
TD-SSP, $S_{\text{stop}} \gg c_t$	Fixed stop + tool-dep.	Lexicographic	Proved
f with C1+C2+C3	General	Yes	Proved (sufficient)